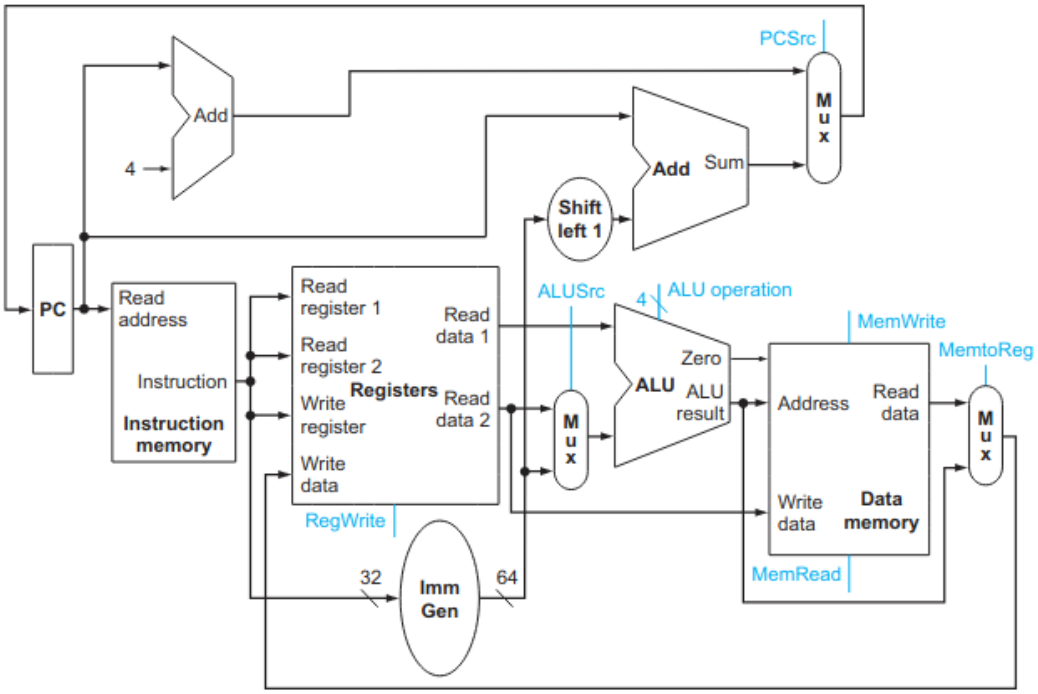
Lab 11: Design and FPGA Implementation of a Single-Cycle RISC-V Processor

Objective

To integrate all previously designed processor components into a complete single-cycle RISC-V (RV32I) processor. Students will design the Instruction Memory, connect all datapath modules, verify the processor through simulation, and deploy it on an FPGA. The final processor will execute a given RISC-V program stored in instruction memory, producing observable results via LEDs or seven-segment displays.



Task 1: Program Counter, Immediate Generator and Branch Infrastructure

Objective

To design and verify the Program Counter update mechanism and immediate generation logic required for sequential execution and branch control in the RISC-V processor.

System Overview

In a single-cycle RISC-V processor, the Program Counter (PC) must update on every clock cycle to either:

- The next sequential instruction address (PC + 4), or
- A branch target address (PC + imm $\ll$ 1) when a branch is taken.

To support this functionality, the processor requires the following infrastructure modules.

Required Modules

Module	Function
ProgramCounter	Stores and updates the PC value
pcAdder	Computes PC + 4
branchAdder	Computes PC + (imm $\ll$ 1)
mux2	Selects next PC value
immGen	Generates I, S and B type immediates

Module Requirements

- PC must update on every positive clock edge.
- pcAdder must add a constant value of 4 to the PC.
- branchAdder must compute PC + (sign-extended immediate $\ll$ 1).
- mux2 must select between sequential and branch target addresses using PCSrc.
- immGen must support I, S and B type immediate formats.

Procedure

(a) Create the modules: `ProgramCounter.v`, `pcAdder.v`, `branchAdder.v`, `mux2.v`, and `immGen.v`.

(b) Implement immediate generation according to the RV32I encoding format.

(c) Connect all modules to form the PC update datapath.

(d) Write a testbench to verify:

- PC increments correctly by 4 when PCSrc = 0.
- PC updates to branch target when PCSrc = 1.
- Immediate generation produces correct signed values.

Deliverables

- Verilog files: `ProgramCounter.v`, `pcAdder.v`, `branchAdder.v`, `mux2.v`, `immGen.v`
- Testbench waveform showing correct PC update and immediate generation

Task 2: Datapath Integration and Top-Level Processor

Objective

To integrate the instruction memory with the remaining datapath components to form the complete single-cycle RISC-V processor. Students will connect PC, control logic, register file, ALU, immediate generator, data memory, and program flow components into one unified design.

Procedure

(a) Instantiate all modules inside a new file named `TopLevelProcessor.v`.

(b) Connect instruction memory output to the control unit and register file inputs.

(c) Connect control signals to ALU, ALU control, register file, and data memory.

(d) Verify all datapath signals match the RISC-V single-cycle architecture.

Deliverables

- Verilog file: `TopLevelProcessor.v`
- Complete datapath diagram schematic
- Utilization report
- Timing Analysis report

Assessment Rubric
Lab 11: Design and FPGA Implementation of a Single-Cycle RISC-V Processor

Name:	Student ID:	Section:

Points Distribution

	Task No.	OE2 (Design)	OE5 (Report)	OE6 (Reflection)
In-Lab	Task 1	/40	-	
	Task 2	-	/30	-
	Viva	-	-	/10
Total Points		**/40**	**/30**	**/10**
CLO Mapped		CLO 3		

Affective Domain Rubric		Points	CLO Mapped
AR7	**Report Submission & Git Upload**	/10 & /10	CLO 3

CLO	Total Points	Points Obtained
3	100	
Total	**100**	

For description of different levels of the mapped rubrics, please refer to the Open Ended Lab rubrics provided here.